

1.3 - Wake Functions

- The ME for the fields generated by a source particle of transv. coord.'s (x_0, y_0) and velocity v along the longitudinal direction z
 - + Boundary Conditions of continuity for $D^\perp, B^\perp, \vec{E}^\parallel$ and $\vec{H}^\parallel$
 - + Expression for momentum change of the witness particle (determ. using the mentioned approximations)
- } $\Rightarrow$ Fields $\vec{E}$ and $\vec{B}$ from the passage of the source part. are uniquely determined

$$\Delta \vec{p}(\vec{p}_s, \vec{p}_w, z) = \int_{-\infty}^{\infty} dt \vec{F}(\vec{p}_s, \vec{p}_w, s, t) \Big|_{s=vt-z}$$

$\vec{E} = \vec{E}(\vec{p}_w, vt-z)$
 $\vec{B} = \vec{B}(\vec{p}_w, vt-z)$

where: $\vec{p}_s = (x_s, y_s)$, $\vec{p}_w = (x_w, y_w)$, $\vec{F} = q \left[E_s \hat{s} + (E_x + v B_y) \hat{x} + (E_y - v B_x) \hat{y} \right]$

$\nearrow$ Lorentz Force

In theory:

$\Rightarrow$ Everything we need to compute the total momentum variation of each witness particle due to the effect of all other (source) particles!

Nonetheless, we can use some additional definitions that will be very handy when one seeks to indeed perform such calculations (and explore symmetries / simplifications).

The first of them are the wake functions (or wakes):

$$(w_x, w_y, w_{||}) = \frac{v}{qQ} (\Delta p_x, \Delta p_y, -\Delta p_s) \sim [V/c] = [J/c^2]$$

$\nearrow w_s > 0 \Rightarrow \text{energy loss}^*$

functions independent of the charges involved!

$$q_w \sim q$$

$$q_s \sim Q$$

w_i 's depend only on the "structure" that generates the wake fields

$$\sim m v^2 \quad [kg \cdot m^2/s^2]$$

$$\sim m \cdot a \cdot d \quad [kg \cdot m^2/s^2]$$

$$p \cdot v \sim F \cdot d$$

1.4 - Wake Potential

Then, having the wake functions $\vec{w}$, one elegant (and powerful) further step is showing that these functions can be obtained from the derivatives of a scalar potential function W , which we'll call wake potential

REMINDER

In electromag.: $\vec{E} \rightarrow$ irrotational field $\Rightarrow \oint \vec{E} \cdot d\vec{l} = 0 \Rightarrow \int_A^B \vec{E} \cdot d\vec{l}$ is path independent

$\rightarrow$ enables defining $V(\vec{r}) \stackrel{\Delta}{=} - \int_O^{\vec{r}} \vec{E} \cdot d\vec{l} \Rightarrow - \int_A^B \vec{E} \cdot d\vec{l} = V_B - V_A$

$\Rightarrow \vec{E}(\vec{r}) = - \vec{\nabla} V(\vec{r})$ or simply $\vec{E} = - \vec{\nabla} V$ $\rightarrow$ gradient of a scalar potential

$\hookrightarrow \sim E_u(\vec{r}) = - \frac{\partial V}{\partial u}(\vec{r})$ $\hookrightarrow$ vector field

And just noticing:

$$E_u = \frac{F_u}{q} = \frac{1}{q} \frac{dp_u}{dt}; \quad w_u \sim \frac{v}{qQ} (\pm) dp_u$$

We want to find a $W(\vec{p}_s, \vec{p}_w, z)$ such that $w_u \propto \frac{\partial W}{\partial u}$

In particular, we'll have: $-\frac{\partial W}{\partial s} \sim$ (the minus sign being due to $w_{||} = -C \cdot \Delta p_s$, $C > 0$)

$$w_{||} = - \frac{\partial W}{\partial (-z)} = \frac{\partial W}{\partial z}; \quad \vec{w}_{\perp} = \vec{\nabla}_{w,\perp} W = \left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w} \right) \cdot W$$

To find such potential, we start by writing the Lagrangian of the problem and, from the Euler-Lagrange equation $\left(\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) = \vec{\nabla} L \right)$, obtain an expression for $\Delta \vec{p}$ ("a" $\vec{w}$), which will, in this specific case, be proportional to the gradient of a scalar function!

$\rightarrow$ the "a" are due to the minus sign in $w_{||} = -\Delta p_s$, so $\vec{w} \neq a \cdot \Delta \vec{p}$, meaning $\vec{w}$ and $\Delta \vec{p}$ are not literally proportional

Reminder on Lagrangian Mechanics

definition in classical mechanics

We begin with the Lagrangian: $L = T - V$

From which we can define the action:

$$S = \int L dt$$

Then, from the Principle of Least Action, we

know that any system will evolve in a way which extremizes $S \sim \delta S = 0$!

But what does $\delta S = 0$ mean? Say we begin at the optimal path $(q, \dot{q})$, so that S_i is the "extreme-valued" action. Then, when making a small deviation $(\delta q, \delta \dot{q})$ from that path, we should find $S_f = S_i + \delta S = S_i$,

i.e., $\delta S = 0$:

$$\begin{aligned} S(q + \delta q, \dot{q} + \delta \dot{q}) &= \int_{t_0}^{t_1} L(q + \delta q, \dot{q} + \delta \dot{q}, t) dt = \int_{t_0}^{t_1} \left[L(q, \dot{q}, t) + \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt = \\ &= S(q, \dot{q}, t) + \int_{t_0}^{t_1} \frac{\partial L}{\partial q} \delta q dt + \int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} \delta q dt \\ &= S(q, \dot{q}, t) + \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q dt \end{aligned}$$

1st order δq 's are small

$\delta \dot{q} = \delta \frac{dq}{dt} = \frac{d}{dt} \delta q$

$\delta q(t_0) = \delta q(t_1) = 0$

$\delta \dot{q} = \dot{q}_f - \dot{q}_i = \frac{d}{dt} (q_f - q_i) = \frac{d}{dt} \delta q$

$\Rightarrow \delta S = S(q + \delta q, \dot{q} + \delta \dot{q}) - S(q, \dot{q}, t) = \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q dt$

If instead of $L(q, \dot{q}, t)$ we had $L(q_i, \dot{q}_i, t)$, we would have $\delta L = \sum_i \frac{\partial L}{\partial q_i} \delta q_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i$, and thus

$$\delta S = \int_{t_0}^{t_1} \delta L(q_i, \dot{q}_i, t) dt = \sum_i \left[\int_{t_0}^{t_1} \frac{\partial L}{\partial q_i} \delta q_i dt + \int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i dt \right] = \int_{t_0}^{t_1} \sum_i \left[\left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right) \delta q_i \right] dt = 0$$

Euler-Lagrange equation *

Now, to have $\delta S = 0$ for any small δq_i , the

remaining of the integrands must all be zero everywhere, and thus

↳ independently

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

V:

• Thus, by building the Lagrangian $L(\vec{q}, \dot{\vec{q}}, t)$ of a physical system, we can determine how $(\vec{q}, \dot{\vec{q}})'$ will evolve - i.e., the equations of movement of the system - by using $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i}$.

• Illustrating the case for classical mechanics (1 particle):

$$\triangleright T = \frac{1}{2} m |\dot{\vec{q}}|^2 \Rightarrow L(\vec{q}, \dot{\vec{q}}, t) = \frac{1}{2} m |\dot{\vec{q}}|^2 - V(\vec{q})$$

$\nearrow = \frac{1}{2} m (\dot{q}_x^2 + \dot{q}_y^2 + \dot{q}_z^2)$

Each coord. u :

$$\triangleright \text{Euler-Lagrange: } \frac{\partial L}{\partial \dot{q}_u} = m \dot{q}_u \Rightarrow \frac{d}{dt} (m \dot{q}_u) = \frac{\partial}{\partial q_u} (-V(\vec{q})) \Rightarrow m \ddot{q}_u = - \frac{\partial}{\partial q_u} V(\vec{q}) = F_u \quad ! \quad \sim \text{Newton's and Law}$$

Or in vec. terms:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\vec{q}}} = \frac{\partial L}{\partial \vec{q}} = \frac{d}{dt} m \dot{\vec{q}} - \vec{\nabla} V \Rightarrow m \ddot{\vec{q}} = - \vec{\nabla} V \quad \sim \left(m \ddot{q}_x, m \ddot{q}_y, m \ddot{q}_z \right) = - \left(\frac{\partial V}{\partial q_x}, \frac{\partial V}{\partial q_y}, \frac{\partial V}{\partial q_z} \right) =$$

→ Thus recovering the movement equation from Newtonian Mechanics!

$$= - \left(\frac{\partial}{\partial q_x}, \frac{\partial}{\partial q_y}, \frac{\partial}{\partial q_z} \right) V = - \vec{\nabla} V$$

• Expressing the E.L. equations for 3 coordinates (and using $\dot{\vec{q}} = \vec{v}$):

$$\boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) = \vec{\nabla} L} \quad \rightarrow \quad \text{If } L = T - V: \quad \frac{d}{dt} (m \vec{v}) = \vec{\nabla} (-V) \Rightarrow \frac{d\vec{p}}{dt} = - \vec{\nabla} V \Rightarrow \frac{d\vec{p}}{dt} = \vec{F}$$

Newton's and Law very directly!

Now that this is clear, we can go back to the wake-fields context and find the Lagrangian of the witness particle (in whose $\Delta \vec{p} = "a" \vec{w}$ - we're interested in).

It is the Lagrangian of a relativistic charged particle moving through E.M. fields $\vec{E}$ and $\vec{B}$, i.e., under the influence of E.M. potentials ϕ and $\vec{A}$.

Let's build it in two parts:

- Relativistic free-particle term
- Electromagnetic interaction term



APPENDIX

(i) Lagrangian of a Relativistic Free-Particle

• In non-relativistic mechanics, we would have simply: $L = T = \frac{1}{2}mv^2$

• But in the relativistic context, we also have the requirement of $\int L dt$ being Lorentz invariant! ~ which is not the case for dt , ds or v ($\neq$ observers can disagree about them).

→ Thus, we should have something with the units of action (energy · time) which is also Lorentz invariant.

↳ The natural invariant along a particle's path is its proper time $d\tau$
 actual time experienced by the particle
 independent of the observer's ref. frame
 $d\tau = \frac{dt}{\gamma}$
 time interval measured by a stationary observer's reference frame
 fixed by the geom. of spacetime
 time registered by a clock which is at rest relative to the particle moving with

→ Then, to adjust for the units,

the simplest L.-inv. term with energy units

would be $A \cdot mc^2 \sim$ prop. to rest mass:

$$L = A mc^2 \sqrt{1 - v^2/c^2}$$

$$\Rightarrow \int L dt = \int A mc^2 \sqrt{1 - v^2/c^2} dt$$

Finally, to determ. A we can use Lagrangian formalism,

from which the canonical momentum is:

$$\vec{p}_{can} = \frac{\partial L}{\partial \vec{v}} = \frac{\partial}{\partial \vec{v}} \left(A mc^2 \sqrt{1 - v^2/c^2} \right) = A mc^2 \cdot \frac{-2\vec{v}/c^2}{2\sqrt{1 - v^2/c^2}} = -A \gamma m \vec{v}$$

Thus, for it to match the relativistic momentum, $\vec{p}_{rel} = \gamma m \vec{v}$, we need $A = -1$

And therefore, for a rel. free-particle the Lagrangian is:

$$L_{rel} = -\frac{mc^2}{\gamma} = -mc^2 \sqrt{1 - v^2/c^2}$$

and thus: $L_{rel} \stackrel{v \ll c}{\approx} -mc^2 \left(1 - \frac{v^2}{2c^2} \right) = -mc^2 + \frac{1}{2}mv^2$ ✓

~ Notice that if $v \ll c \Rightarrow \frac{v^2}{c^2} \ll 1$:

$$(1 - u)^{1/2} = 1 + \frac{1}{2}(1 - u)^{-1/2} \cdot u + \dots \stackrel{u \ll 1}{\approx} 1 - \frac{u}{2}$$

at $u=0$
 $\Rightarrow \frac{\partial L}{\partial \vec{v}} = m \vec{v} = \vec{p}_{classical}$
 i.e. doesn't affect the eq's of motion:

(ii) Electromagnetic Interaction Lagrangian Term

• Forgetting relativity for a moment, ^(v < c) we want to write a Lagrangian for which the Euler-Lagrange eq. gives a force term which is precisely the Lorentz Force, $\vec{F}_L = q(\vec{E} + \vec{v} \times \vec{B})$.

• We remember that $\vec{E}$ and $\vec{B}$ (due to satisfying $\vec{\nabla} \times \vec{E} = 0$ and $\vec{\nabla} \cdot \vec{B} = 0$) can be written in terms of the electric and magnetic potentials: $\vec{E} = -\vec{\nabla}\phi$, $\vec{B} = \vec{\nabla} \times \vec{A}$.
 $\Rightarrow \vec{E} = -\vec{\nabla}\phi \sim$ "electrostatic case"
 $\phi = \phi(\vec{r}) \sim \text{scal. 3D field}$
 $\vec{A} = \vec{A}(\vec{r}) \sim \text{vec. 3D field}$

• To build an intuition, we start with the case where only an electric force is acting on the particle (purely electrostatic $\sim \vec{B} = 0 \Rightarrow \vec{E} = -\vec{\nabla}\phi$)

$\rightarrow$ The potential energy is then: $V_{el} = q \cdot \phi(\vec{r})$
 Euler-Lagrange: $\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) = \vec{\nabla} V$

$\rightarrow$ Plugging into the Lagrangian: $L = \frac{1}{2} m |\vec{v}|^2 - q\phi$
 $\Rightarrow \frac{d}{dt} \vec{p} = -\vec{\nabla} q\phi \Rightarrow \vec{F}_E = -\vec{\nabla} V_{el}$ ✓

From now on we should make the Maxwell correction $\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$

• Now, to account for the $q\vec{v} \times \vec{B}$ of the Lorentz force, we see that

$\vec{v} \times \vec{B} = \vec{v} \times \vec{\nabla} \times \vec{A} \stackrel{\text{B(A.C) - C(A.B) rule w/ operator abstractions for and (but)}}{=} \underbrace{(\vec{\nabla}(\vec{v} \cdot \vec{A}) - \vec{A}(\vec{\nabla} \cdot \vec{v}))}_{\text{and based on the 1st term, make the guess:}}$

$\rightarrow V = q\phi - q\vec{v} \cdot \vec{A}$

$\rightarrow$ Lagrangian: $L = \frac{1}{2} m |\vec{v}|^2 - q\phi + q\vec{v} \cdot \vec{A}$
 E.L.: $\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) - \vec{\nabla} L = \frac{d}{dt} (m\vec{v} + q\vec{A}) - q\vec{\nabla}(-\phi + \vec{v} \cdot \vec{A}) =$
 $= \frac{d\vec{p}}{dt} + q \frac{d\vec{A}}{dt} + q\vec{\nabla}\phi - q\vec{\nabla}(\vec{v} \cdot \vec{A}) = 0$ *

But notice that $\frac{d}{dt} \vec{A}(\vec{r}) = \frac{\partial \vec{A}}{\partial t} + \frac{\vec{r}}{\partial t} \cdot \frac{\partial \vec{A}}{\partial \vec{r}} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{A}$
 $\sim \frac{d\vec{A}}{dt}(\vec{r}, t) = \frac{\partial A_x}{\partial t} + \left(\frac{\partial A_x}{\partial x} \frac{dx}{dt} + \frac{\partial A_x}{\partial y} \frac{dy}{dt} + \frac{\partial A_x}{\partial z} \frac{dz}{dt} \right) = \frac{\partial A_x}{\partial t} + (\vec{v} \cdot \vec{\nabla}) A_x$

And thus:

* $\Rightarrow \frac{d\vec{p}}{dt} = +q \left[\underbrace{\frac{\partial \vec{A}}{\partial t} - (\vec{v} \cdot \vec{\nabla}) \vec{A}}_{= \vec{\nabla} \times (\vec{v} \times \vec{A})!} + \vec{\nabla}(\vec{v} \cdot \vec{A}) - \vec{\nabla}\phi \right] = q \left[\underbrace{\left(\vec{\nabla}\phi + \frac{\partial \vec{A}}{\partial t} \right)}_{\vec{E}} + \underbrace{\vec{v} \times (\vec{\nabla} \times \vec{A})}_{\vec{v} \times \vec{B}} \right]$

$\Rightarrow \frac{d\vec{p}}{dt} = q(\vec{E} + \vec{v} \times \vec{B}) = \vec{F}_L$

$\sim$ properly giving Newton's 2nd Law for a non-relativistic charged particle w/ velocity $\vec{v}$ under effect of the Lorentz force. ✓

Thus, the full Lagrangian for the witness particle (relativistic charged particle in electromagnetic potentials ϕ and A) is:

$$L = -mc^2 \sqrt{1 - v^2/c^2} + q \vec{A} \cdot \vec{v} - q\phi, \quad (I)$$

such that:

$$S = \int L dt = \underbrace{-mc^2 \int \frac{dt}{\gamma}}_{S_{rel}} + \underbrace{q \int (\vec{A} \cdot \vec{v} - \phi) dt}_{S_{int.}}$$

$\bullet dS_{rel} = -mc^2 d\tau$
 $\sim$ a particle with mass accumulates action proportional to its proper time $d\tau$
 $\bullet dS_{int.} = q(\vec{A} \cdot d\vec{r} - \phi dt)$
 $\sim$ a charged particle accumulates action by moving through electromagnetic potentials

where $\vec{A}$ and ϕ are the electromag. potentials defined by the ME for the source particle together with the definitions $\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$ and $\vec{B} = \vec{\nabla} \times \vec{A}$.
 which gives $\vec{E}$ and $\vec{B}$

Then, applying the Euler-Lagrange equations to (I):

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) - \vec{\nabla} L = \frac{d}{dt} \left(-mc^2 \frac{-\vec{v}/c^2}{\sqrt{1-v^2/c^2}} + q\vec{A} \right) - q\vec{\nabla} (\vec{A} \cdot \vec{v} - \phi) = \frac{d}{dt} (\gamma m \vec{v} + q\vec{A}) - q\vec{\nabla} (A_z v_z - \phi)$$

$v_x, v_y \ll v_z$

$$\Rightarrow \frac{d}{dt} (\vec{p}_{rel} + q\vec{A}) = q\vec{\nabla} (A_z v_z - \phi)$$

Integrating both sides:

$$\left[\vec{p} + q\vec{A} \right]_{t=-\infty}^{t=+\infty} = \underbrace{\Delta \vec{p}}_{\substack{\text{momentum} \\ \text{change}}} = q \cdot \int_{-\infty}^{+\infty} \vec{\nabla} (A_z v_z - \phi) dt = q \vec{\nabla} \int_{-\infty}^{+\infty} (A_z v_z - \phi) dt$$

assumption that x, y are fixed with t
 gradient of a scalar function
 as we derived!
 $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$

Remembering that $\vec{W} = \frac{c}{qQ} (\Delta p_x, \Delta p_y, -\Delta p_z)$ and seeing that we can write $\Delta \vec{p} = \frac{qQ}{c} \vec{\nabla}_R W$, where

$$\vec{R} = (x_w, y_w, -z) \Rightarrow \Delta \vec{p} = \frac{qQ}{c} \left(\frac{\partial W}{\partial x_w}, \frac{\partial W}{\partial y_w}, -\frac{\partial W}{\partial z} \right), \text{ we have:}$$

$$\Rightarrow (w_x, w_y, w_z) = \frac{c}{qQ} \cdot \frac{qQ}{c} \left(\frac{\partial W}{\partial x_w}, \frac{\partial W}{\partial y_w}, -\frac{\partial W}{\partial z} \right) \Rightarrow \begin{cases} w_x = \frac{\partial W}{\partial x_w} \\ w_y = \frac{\partial W}{\partial y_w} \\ w_z = -\frac{\partial W}{\partial z} \end{cases} !$$

So, in short:

$$\Delta \vec{p} = \vec{\nabla} \left(\underbrace{\frac{q}{c} \int_{-\infty}^{\infty} |A_s v_s - \phi| dt}_{f(\vec{r})} \right) = \frac{q \cdot Q}{c} (w_x, w_y, -w_z)$$

$\text{kg m/s} \quad \text{kg m}^2/\text{s}^2 \cdot 1/c^2 = \mathcal{T}/c^2 = V/c$

$$\Rightarrow (w_x, w_y, -w_z) = \frac{c}{qQ} \cdot \vec{\nabla} f = \vec{\nabla} \left(\frac{c}{qQ} \int_{-\infty}^{\infty} |A_s v_s - \phi| dt \right) = \vec{\nabla}_R W(\vec{\rho}_s, \vec{\rho}_w, z)$$

$\left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial(z-z)} \right)$
 $\sim z$

And thus:

$$(w_x, w_y, -w_z) = \vec{\nabla}_R W \Rightarrow \begin{cases} (w_x, w_y) = \vec{w}_\perp = \vec{\nabla}_{w,\perp} W \\ w_z = -\frac{\partial}{\partial z} W = \frac{\partial}{\partial z} W \end{cases}$$

Furthermore, this potential formalism enables directly obtaining the so-called Panofsky-Wenzel theorem from the second cross derivatives of W :

$$\frac{\partial \vec{w}_\perp}{\partial z} = \frac{\partial}{\partial z} (\vec{\nabla}_{w,\perp} W) = \vec{\nabla}_{w,\perp} \left(\frac{\partial}{\partial z} W \right) = \vec{\nabla}_{w,\perp} w_z \rightarrow \frac{\partial \vec{w}_\perp}{\partial z} = \vec{\nabla}_{w,\perp} w_z \quad (\text{II})$$

↳ This property of the wakes is very useful as it enables calculating $\vec{w}_\perp$ from w_z by

a simple integration of (II):

$$\vec{w}_\perp = \int_{-\infty}^z \vec{\nabla}_{w,\perp} w_z dz^* \quad \text{Furthermore, } w_z = -\frac{c}{qQ} \Delta p_s = -\frac{c}{qQ} \int_{-\infty}^{\infty} \underbrace{(F_L)}_{\text{impulse approx.}} dt = -\frac{c}{Q} \int_{-\infty}^{\infty} dt E_s|_{s=ct-z}$$

$\vec{F}_L = q[E_s \hat{s} + (\epsilon_x - vB_y)\hat{x} + (\epsilon_y + vB_x)\hat{y}]$

⇒ All wake functions can be calculated from E_s !

$$w_z(\vec{\rho}_s, \vec{\rho}_w, z) = -\frac{c}{Q} \int_{-\infty}^{\infty} dt E_s(\vec{\rho}_s, \vec{\rho}_w, z, t) \Big|_{s=ct-z}; \quad \vec{w}_\perp = \int_{-\infty}^z \vec{\nabla}_{w,\perp} w_z dz^* !$$

Another important property of the wake potential W is being an harmonic function in the transverse coordinates x and y [if $\beta = 1 \sim v = \beta c$]*:

$$\nabla_{\perp}^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0 \quad \sim \quad \frac{\partial w_x}{\partial x} = -\frac{\partial w_y}{\partial y}$$

To prove this, we first remember that, from the ME in free space, the potentials ϕ and $\vec{A}$ satisfy the wave equation:

$$\nabla^2 \vec{A} - \overset{1/c^2}{\epsilon_0} \frac{\partial^2 \vec{A}}{\partial t^2} = 0; \quad \nabla^2 \phi - \epsilon_0 \frac{\partial^2 \phi}{\partial t^2} = 0$$

Now, since $W = q \int_{-\infty}^{\infty} (A_s v - \phi) dt$, we have:

$$\nabla^2 W = \nabla_{\perp}^2 W + \frac{\partial^2 W}{\partial z^2} = \nabla^2 \left(q \int_{-\infty}^{\infty} (A_s v - \phi) dt \right) = q \int_{-\infty}^{\infty} (c \nabla^2 A_s - \nabla^2 \phi) dt = q \int_{-\infty}^{\infty} \frac{1}{c^2} \frac{\partial^2}{\partial t^2} (v A_s - \phi) dt$$

$$\Rightarrow \nabla_{\perp}^2 W = q \int_{-\infty}^{\infty} \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) (v A_s - \phi) dt$$

$$\text{Furthermore, } \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) = \left(\frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right) \cdot \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right), \text{ and } z = \beta c t - z \Rightarrow \frac{d}{dt} = \left(\frac{\partial}{\partial t} + \frac{\partial z}{\partial t} \frac{\partial}{\partial z} \right) = \beta c \left(\frac{1}{\beta c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right),$$

meaning that, for $\beta = 1$:

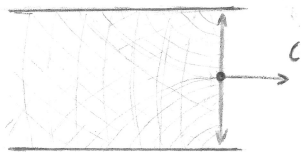
$$\nabla_{\perp}^2 W = q \int_{-\infty}^{\infty} c \frac{d}{dt} \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) (c A_s - \phi) dt = q c \cdot \left[\left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) (c A_s - \phi) \right]_{t=-\infty}^{t=\infty} = 0 \quad \sim \text{since } \vec{A} \text{ and } \phi \text{ vanish at } \pm \infty$$

$$\therefore \nabla_{\perp}^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0$$

1.5 - The "Causality" Principle

Related to this discussion is the remark that, in the ultra-relativistic limit ($v \rightarrow c$), the wakes generated by a particle can only influence particles that are behind it*, since the EM fields (and thus the "wake wave-front") also propagate at speed c .

Wake analogy:



This is equivalent to saying that W must satisfy:

$$W(\vec{p}_s, \vec{p}_w, z) = 0 \quad \forall \quad z < 0 \quad !$$

Which also implies:

$$\left\{ \begin{array}{l} \bullet \vec{w}_\perp - \vec{\nabla}_{w,\perp} W = 0 \quad \forall \quad z < 0 \\ \bullet w_{||} = \frac{\partial W}{\partial z} = 0 \quad \forall \quad z < 0 \end{array} \right.$$

Even though this is a good approx. for real beams in SL storage rings, the raw effective wakes from time-domain solvers do not respect it, as they are calculated from finite-length charge distributions

The wake formalism also applies to the modelling of other sources of col. effects, such as the interaction of the beam w/ ions in the vacuum chamber, and to model the effect of CSR on the beam ~ former $\rightarrow$ causal $\checkmark$
latter $\rightarrow$ not causal !